

MAT1100 Homework 1

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Question 2

Let $G \curvearrowright X \neq \emptyset$ and let G_x denote the stabilizer of $x \in X$. Let $\rho : G \rightarrow \text{Sym}(X)$ denote the permutation representation of the action.

(a) *Proof.* Suppose $g \in G$ and $x, y \in X$ such that $g \cdot x = y$. Then clearly $x = g^{-1} \cdot y$.

($\subseteq$) Suppose $h \in G_y$ it suffices to show $h \in gG_xg^{-1}$ and indeed

$$g^{-1}hg \cdot x = g^{-1}h \cdot y = g^{-1} \cdot y = x$$

hence $g^{-1}hg \in G_x$ and therefore $G_y \subseteq gG_xg^{-1}$

($\supseteq$) Suppose $h \in G_x$ then

$$ghg^{-1} \cdot y = gh \cdot x = g \cdot x = y$$

hence $ghg^{-1} \in G_y$ and therefore $gG_xg^{-1} \subseteq G_y$.

□

(b) *Proof.* First we show that $\ker \rho = \bigcap_{x \in X} G_x$.

($\subseteq$) If $g \in \ker \rho$ then for all $x \in X$, $g \cdot x = x$. Hence $g \in G_x$ for all $x \in X$ and so

$$g \in \bigcap_{x \in X} G_x$$

($\supseteq$) If $g \in \bigcap_{x \in X} G_x$ then for all $x \in X$ $g \in G_x$ and therefore $g \cdot x = x$ for all $x \in X$. In other words, $g \in \ker \rho$.

If $G \curvearrowright X$ is transitive then given $x \in X$

$$\mathcal{O}_x = \{g \cdot x : g \in G\} = X$$

and by (a), $\mathcal{O}_{gx} = g\mathcal{O}_xg^{-1}$. Thus

$$\ker \rho = \bigcap_{x \in X} G_x = \bigcap_{g \in G} G_{g \cdot x} = \bigcap_{g \in G} gG_xg^{-1}$$

□

(c) *Proof.* Fix $x \in X$. Let $\phi : G \rightarrow X$ be given by

$$\phi(g) = g \cdot x$$

First we show that for all $y \in X$ there exists unique $g \in G$ such that

$$g \cdot x = y$$

the existence of such a G is evident as $G \curvearrowright X$ is transitive. Therefore ϕ is surjective.

Now for uniqueness, suppose $g, h \in G$ such that

$$g \cdot x = h \cdot x$$

then

$$g^{-1}h \cdot x = x$$

hence $g^{-1}h \in G_x$. Now given $a \in G_x$, $gag^{-1} = a$ as G is abelian hence $gG_xg^{-1} = G_x$ for all $g \in G$. Since $G \curvearrowright X$ is transitive

$$\ker \rho = \bigcup_{g \in G} gG_xg^{-1} = \bigcap_{g \in G} G_x = G_x$$

hence $g^{-1}h \in G_x = \ker \rho$ and since ρ is faithful this implies that $g^{-1}h = e$ and $g = h$.

The uniqueness of the above element g in turn implies that ϕ is injective. Therefore ϕ is a bijection. $\square$